14350 Computer Graphics Assignment 2 - 31/10/2025

Loading, Illuminating and Rendering Meshes in $\mathbb{R}^3$

Supervisor: Abel Gomes *Scribe: Abel Gomes*

The goal of this assignment is loading and rendering meshes in in $\mathbb{R}^3$ using modern OpenGL. These meshes (format .OBJ files) constitute the input data for the graphics program. After loading the OBJ mesh in memory, you must be able to rotate and translate in $\mathbb{R}^3$ using the GLFW resources concerning mouse and keyboard. You also need to define a point light in your 3D scene to illuminate it. Use the Phong shading to render your 3D scene. Textures are not mandatory.

1 Reference Code

A similar example is described at: `http://www.opengl-tutorial.org/beginners-tutorials/tutorial-7-model-loading/`. See also the section named Model Loading at `https://learnopengl.com` for an possible option. As a third option, you can use `https://github.com/tinyobjloader`. Besides, have a look at for a better understanding of obj format: `https://www.scratchapixel.com/lessons/3d-basic-rendering/obj-file-format/obj-file-format.html`.

2 Who does what?

The first digit (the rightmost) of the student number identifies the work. See next section to make sure about your assignment.

3 Exercises

0. UFO (`https://www.cgtrader.com/free-3d-models/space/spaceship/free-flying-saucer`).
1. Deers (`https://www.cgtrader.com/free-3d-models/animals/mammal/origami-deer`).
2. Bar stool (`https://www.cgtrader.com/free-3d-models/furniture/chair/simple-barstool`).
3. Boat (`http://surl.li/mqjru`).
4. Cat (`https://www.cgtrader.com/free-3d-models/animals/mammal/base-mesh-cat`).
5. Table (`http://surl.li/mqjsc`).
6. House (`http://surl.li/mqjsi`).
7. Apple (`http://surl.li/mqjsl`).
8. Coca-Cola (`http://surl.li/mqjsr`).
9. Wall clock ()`https://www.cgtrader.com/free-3d-models/furniture/other/p-c-n`).

4 Deadline

The deadline to submit this programming assignment is November 30, 2025, 23:59.

References

[1] OBJ format wiki: `https://en.wikipedia.org/wiki/Wavefront_.obj_file`

[2] More notes about OBJ format: `https://www.marxentlabs.com/obj-files/`

[3] The OpenGL Shading Language `https://www.opengl.org/registry/doc/GLSLangSpec.4.40.pdf`, last access on 08/04/2015.

[4] Dave Shreiner, Graham Sellers, John Kessenich, and Bill Licea-Kane. OpenGL Programming Guide, 8th edition, version 4.3. Addison-Wesley, Upper Saddle River, 2013.